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Chapter 1

Variational Inference

Abstract We present here variational inference methods to reason on Computation-Activation Networks.
Topics:

- I.3.4 (Exponential family) Variational inference
- I.3.7 Mean field methods

1.1 Foundational inference problems

We have two parametrizations of maximum entropy distributions:

- **Mean parametrization:** The mean parameter $\mu [L] \in \mathcal{M}_{t,\nu}$
- **Canonical parametrization:** The face $\mathcal{F} \in \mathcal{Q}_{t,\nu}^I$ together with the weight vector $\theta [L]$

The fundamental inference problem are finding one parametrization given the other.

- **Forward mapping:** Given the canonical parametrization $(\mathcal{F}, \theta [L])$, find the mean parameter $\mu [L]$.
- **Backward mapping:** Given the mean parameter $\mu [L]$, find the face $\mathcal{F}$ such that $\mu [L]$ is in the relative interior of the face, and the corresponding canonical parameter $\theta [L]$.

1.1.1 Forward mapping

The forward mapping is the contraction of the distribution with the selection encoding of the statistic, that is the map

$$F^{(t,\nu)}(t) : \mathcal{P}_t \rightarrow \mathcal{M}_t$$

with

$$F^t(\mathcal{F}, \theta) = \frac{\langle \alpha^\theta [Y_{[p]}], \kappa^\mathcal{F} [Y_{[p]}], \beta^t [Y_{[p]}, X_{[d]}], \sigma^t [X_{[d]}, L] \rangle_{[L]}}{\langle \alpha^\theta [Y_{[p]}], \kappa^\mathcal{F} [Y_{[p]}], \beta^t [Y_{[p]}, X_{[d]}] \rangle_{[\emptyset]}}.$$

1.1.2 Backward mapping

Determining the face such that $\mu [L]$ in the relative interior, and the corresponding canonical parameter.

1.1.2.1 Cross entropy minimization

The cross entropy between a distribution $\mathbb{P}^* [X_{[d]}]$ and a maximum entropy distribution $\mathbb{P}^{\mathcal{F}, \theta} [X_{[d]}]$ is given by

$$\begin{aligned} \mathbb{H} [\mathbb{P}^* [X_{[d]}], \mathbb{P}^{\mathcal{F}, \theta} [X_{[d]}]] &= - \langle \mathbb{P}^* [X_{[d]}], \beta^t [Y_{[p]}, X_{[d]}], \ln [\alpha^\theta [Y_{[p]}]] \rangle_{[\emptyset]} \\ &\quad - \langle \mathbb{P}^* [X_{[d]}], \beta^t [Y_{[p]}, X_{[d]}], \ln [\kappa^{\mathcal{F}} [Y_{[p]}]] \rangle_{[\emptyset]} \\ &\quad + \ln \left[\langle \alpha^\theta [Y_{[p]}], \kappa^{\mathcal{F}} [Y_{[p]}], \beta^t [Y_{[p]}, X_{[d]}] \rangle_{[\emptyset]} \right] \end{aligned}$$

The first term can be simplified as

$$\begin{aligned} - \langle \mathbb{P}^* [X_{[d]}], \beta^t [Y_{[p]}, X_{[d]}], \ln [\alpha^\theta [Y_{[p]}]] \rangle_{[\emptyset]} &= - \sum_{\ell \in [p]} \theta [L = \ell] \cdot \langle \mathbb{P}^* [X_{[d]}], t_\ell [X_{[d]}] \rangle_{[\emptyset]} \\ &= - \langle \mu^* [L], \theta [L] \rangle_{[\emptyset]} \end{aligned}$$

To investigate the second term, we notice that

$$\langle \beta^t [Y_{[p]}, X_{[d]}], \ln [\kappa^{\mathcal{F}} [Y_{[p]}]] \rangle_{[X_{[d]}=x_{[d]}]} = \begin{cases} 0 & \text{if } \gamma^{x_{[d]}} \in \mathcal{F} \\ -\infty & \text{else} \end{cases}$$

Therefore the cross entropy is finite, if and only if $\mathbb{P}^* [X_{[d]}]$ is supported at the face measure. This is only the case if $\mu^* [L] \in \mathcal{F}$. If the cross entropy is finite, the second term vanishes.

Among the faces with $\mu^* [L] \in \mathcal{F}$ the third term is minimal for each $\theta \in \mathbb{R}^p$.

$$\begin{aligned} &\operatorname{argmin}_{\mathcal{F} \in L(\mathcal{M}_t, \nu)} \operatorname{argmin}_{\theta \in \mathbb{R}^p} \mathbb{H} [\mathbb{P} [X_{[d]}], \mathbb{P}^{\mathcal{F}, \theta} [X_{[d]}]] \\ &= \operatorname{argmin}_{\theta \in \mathbb{R}^p} - \langle \mu^* [L], \theta [L] \rangle_{[\emptyset]} + \ln \left[\langle \alpha^\theta [Y_{[p]}], \kappa^{\mathcal{F}^{\mu^*}} [Y_{[p]}], \beta^t [Y_{[p]}, X_{[d]}] \rangle_{[\emptyset]} \right] \end{aligned}$$

1.1.2.2 Definition of backward mapping

1.2 Face estimation

Generic situation:

Find maximal face in lattice $L(\mathcal{M}_t)$ with

Theorem 1.1 *Let $\mathbb{P}^* [X_{[d]}]$ be a distribution with mean parameter $\mu^* [L]$. Then the face $\mathcal{F}(\mu^*)$ is the maximal face in the lattice $L(\mathcal{M}_t)$ such that*

$$\langle \mathbb{P}^* [X_{[d]}], \beta^t [Y_{[p]}, X_{[d]}], \kappa^{\mathcal{F}} [Y_{[p]}] \rangle_{[\emptyset]} = 1.$$

1.2.1 Cube-like faces

Cube-faces are parametrized by $\mathcal{U}$ and a boolean tuple $y_{\mathcal{U}}$. For cube-like faces, it is enough to decide entailment and contradiction of each feature.

Theorem 1.2 *If the mean parameter $\mu^* [L]$ to a distribution $\mathbb{P}^* [X_{[d]}]$ is in the relative interior of a cube-like face, then the face $\mathcal{F}(\mu^*)$*

$$\mathcal{U} = \{\ell : \ell \in [p], \langle \mathbb{P}^* [X_{[d]}], t_\ell [X_{[d]}] \rangle_{[\emptyset]} \in \{0, 1\}\}$$

and

$$y_\ell = \langle \mathbb{P}^* [X_{[d]}], t_\ell [X_{[d]}] \rangle_{[\emptyset]} .$$

1.3 Parameter estimation

We now present iterative algorithms to train the weight vector, once the face to a mean parameter has been determined. They are coordinate descent methods on the cross-entropy loss function.

1.3.1 Moment matching

We differentiate the cross-entropy with respect to the weight vector

$$\frac{\partial \mathbb{H} [\mathbb{P}^*, \mathbb{P}^{\mathcal{F}}, \theta]}{\partial \theta [L = \ell]} = \mu [L = \ell] - \mu^* [L = \ell] .$$

The derivative vanishes when both mean parameters coincide, which is the moment matching condition.

1.3.2 Issues with erroneous faces

If applied to the wrong face, they fail:

- If $\mu [L]$ not on the face, then the cross-entropy is infinite.
- If $\mu [L]$ on the face, but not in the relative interior, then the norm of the weight vector diverges.